

MuleTrace AI

Complete Technology Stack for Cross-Channel Mule Account Detection & Financial Crime Investigation Platform

Layer	Technologies
Frontend	React 19, TypeScript, Vite, Tailwind CSS, shadcn/ui, Framer Motion, React Router, TanStack Query, Zustand, Axios, Socket.IO Client, React Hook Form, Zod
Visualization	Chart.js, React Flow, React Leaflet, Leaflet, OpenStreetMap, Lucide React, Recharts
Backend	NestJS, TypeScript, Prisma ORM, Socket.IO, Swagger, BullMQ, Redis
AI Service	Python 3.12, FastAPI, PyTorch, PyTorch Geometric, Scikit-learn, NetworkX, XGBoost, SHAP, Pandas, NumPy
AI Agents	Risk Assessment, Graph Intelligence, Geo Intelligence, Investigation, AI Copilot, Fraud Replay, Report Generation
Databases	PostgreSQL (Neon), Neo4j AuraDB, Redis Cloud
Geo Intelligence	React Leaflet, OpenStreetMap, GeoPy, MaxMind GeoLite2, PostGIS (Future)
Graph Intelligence	Neo4j, Cypher, NetworkX, PyTorch Geometric, PageRank, Louvain Community Detection, Shortest Path, Centrality Analysis
Real-Time	Socket.IO, Redis Pub/Sub, BullMQ
Reports	ReportLab, Matplotlib, openpyxl
AI Copilot	Ollama, Qwen 2.5 or Llama 3.1, LangGraph, LangChain
Dataset Generator	Python, Faker, Pandas
Fraud Models	XGBoost, Isolation Forest, Random Forest, Graph Neural Networks
Testing	Jest, Vitest, Playwright
Deployment	Vercel, Railway, Neon PostgreSQL, Neo4j AuraDB, Redis Cloud
DevOps	Docker, Docker Compose, GitHub Actions
Development Tools	VS Code, Git, GitHub, Postman, Neo4j Browser, pgAdmin, Docker Desktop, Figma

Recommended Architecture

React + TypeScript Dashboard → NestJS API → PostgreSQL + Neo4j + Redis → FastAPI AI Service → Risk Assessment Agent → Graph Intelligence Agent → Geo Intelligence Agent → Investigation Agent → AI Copilot → Report Generation.